Design System & Cognitive Architecture

1. Design Philosophy: Cognitive Engineering

This system is built on **Cognitive Load Theory (CLT)**. The user context is high-stakes, real-time interpersonal engagement. The UI must act as an "external brain," managing working memory limitations through specific information organization strategies.

Core Strategies & Implementation

Cognitive Strategy	UI Implementation	Research Basis
Chunk & Prioritize	**Progressive Disclosure:** Accordions for all "Deep Dive" content. **The Rule of 5:** No list view exceeds 5 items without a "Show More" interaction.	Miller's Law (7 ± 2 items)
Graphic Organizers	**Split-Views:** Comparison (Type C) pages use toggle-based views on mobile (side-by-side on tablet) to visually separate worldviews.	Dual Coding Theory
Mental Links	**Semantic Icons:** Consistent iconography (Book = Scripture, Bubble = Language) to trigger schema activation.	Schema Theory
Active Retrieval	**Blur/Reveal Patterns:** Key answers in "Practice" modules are obscured until tapped to force memory recall.	Testing Effect (Roediger/Karpicke)
Contextual Relevance	**Sensitivity Coding:** Color-coded borders indicate the "Social Safety" of a topic (Green=Safe, Amber=Caution).	Yerkes-Dodson Law (Eustress)

2. Visual Hierarchy & Typography (Apple HIG)

We utilize the system font stack (San Francisco on iOS) to leverage user familiarity and legibility.

Typography Scale

Designed for Context-Specific Reading Patterns:

- **F-Pattern (Scanning):** Used for Type A (Overview) and B (Concept) pages.
- **Layer-Cake Pattern (Headings):** Used for Type D (Conversation) pages to allow users to skip to the 'Response' section rapidly.

- **Pinball Pattern (Oscillating):** Anticipated for Type C (Comparison) pages; visual anchors must exist on both sides of the comparison to guide the eye.
- **Page Title (Type A-F):** 28pt / 34pt (Dynamic Type: Large Title). Bold.
  - *Purpose:* Immediate orientation.
- **Section Header (Chunk Labels):** 22pt / 28pt (Title 2). Semibold.
  - *Purpose:* Delimits "chunks" of information.
- **Body Text:** 17pt / 22pt (Body). Regular.
  - *Constraint:* Max line length 60 characters (approx 350px width) for readability.
- **Meta/Tags:** 13pt (Footnote). Gray Secondary.
  - *Purpose:* Contextual metadata (Sensitivity, Depth).

Color System (Semantic & Emotional)

Aligned with `01-language-guardrails.md` —avoiding aggressive/corrective tones.

- **Primary (Action):** `Color-Blue` (#007AFF) - Used for navigation and "Next" actions.
- **Neutral (Background):** `System-Background` (#FFFFFF / #000000) - Maximum contrast.
- **Sensitivity Indicators (Border/Icon Tints):**
  - **Low Sensitivity (Safe):** `Color-Teal` (#30B0C7) - Invites open dialogue.
  - **Medium Sensitivity (Caution):** `Color-Orange` (#FF9500) - Requires nuance.
  - **High Sensitivity (Private):** `Color-Purple` (#AF52DE) - Deep theological divergence. *Avoid Red (danger).*

3. Component Architecture & Usability (NN/g)

A. Navigation (The Thumb Zone)

- **Pattern:** Bottom Tab Bar (Fixed).
- **Items:** Learn, Compare, Practice, Search.
- **GOMS Analysis:** Placing nav at bottom reduces Fitts's Law movement time (T) by 40% compared to a Hamburger menu on modern large phones.

B. The "Accordion Stack" (Chunking)

Used for `03-page-templates.md` Type A and B.

- **Behavior:** Only one section open at a time (auto-collapse others).
- **Visuals:** 1px hairline separator. Chevron indicator on right.

- **Touch Target:** Full width, min-height 44pt (Apple HIG minimum).

C. The "Comparison Matrix" (Graphic Organizer)

Used for `03-page-templates.md` Type C.

- **Mobile State:** Segmented Control (Toggle) at top:

$$ChristianView|MuslimView$$

  . Content changes below without page reload.
- **Desktop/Tablet State:** Two-column grid side-by-side.
- **Rationale:** Reducing visual noise allows the user to focus on one paradigm at a time while keeping the "counterpart" one tap away.

D. The "Conversation Flow" (Logical Sequencing)

Used for `03-page-templates.md` Type D.

- **Visual:** Vertical step-progress line (Stepped Process).
- **Nodes:**
  1. "What you hear" (Quote bubble)
  2. "What they mean" (Text block)
  3. "How to respond" (Action block)
- **Interaction:** Tapping a step highlights it and dims others (Focus Mode).

E. Active Retrieval Cards

Used for Practice & Language Modules.

- **State:** Default = Obscured (Blurred text or "Tap to Reveal" overlay).
- **Interaction:** Tap to reveal.
- **Feedback:** "Got it" (Green check) vs "Review" (Refresh icon).

4. Usability Guardrails (KLM-GOMS Optimized)

To ensure the app performs under pressure, we optimize for **Keystroke-Level Model (KLM)** efficiency.

Action	Target Time	Design Requirement

Find a Phrase	< 5 sec	**Search:** Global search bar always visible in "Language" tab. Auto-focus on tap. Search algorithm prioritizes "Tags" over body text.
Switch Context	< 1 sec	**State Preservation:** If user leaves "Compare" tab to check "Scripture", returning to "Compare" must restore exact scroll position.
Read Logic	N/A	**Visual Anchors:** Every page Type D (Conversation) must start with a "Quick Summary" box (grey background) before the details.

5. Accessibility & Inclusivity

- **Contrast:** All text must meet WCAG AA (4.5:1).
- **Touch Targets:** Minimum 44x44pt for all interactive elements.
- **Dynamic Type:** Layouts must not break at 200% font size.
- **Dark Mode:** Mandatory. Reduces eye strain in low-light evening ministry contexts.

6. Implementation Checklist for Devs

1. **Set up CSS Variables** for Sensitivity Colors (Teal, Orange, Purple).
2. **Build the** `Accordion` **component** with `details/summary` HTML tags for native accessibility.
3. **Implement** `SegmentedControl` for the Type C Comparison views.
4. **Create** `RevealCard` **component** for the Practice section.